Tharun Kumar Aduri
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PROFESSIONAL SUMMARY:

Accomplished and results-driven Lead Data Engineer with over 11+ years of experience in architecting, developing, and optimizing scalable, cloud-native data solutions across AWS and Azure platforms. Expertise in building complex ETL pipelines, orchestrating large-scale data migrations, and implementing both batch and real-time data processing using tools like Apache Airflow, AWS Glue, Informatica, and Azure Data Factory, with strong programming skills in Python and PySpark.

Proficient in modern data warehousing and analytics platforms including Amazon Redshift, Snowflake, Azure Synapse, and Azure Data Lake, supporting business-critical reporting, compliance, and advanced analytics. Hands-on experience with real-time streaming using Kafka and Amazon Kinesis, and integrating data from APIs, web sources, and third-party platforms such as CM, DV360, and Search Ads.

Strong background in applying AI/ML techniques for data quality monitoring, predictive modeling, and campaign optimization. Collaborated closely with data scientists to develop, test, and deploy machine learning models using tools such as Amazon SageMaker and Databricks, contributing to intelligent automation and business insights at scale.

Experienced in building and maintaining containerized CI/CD environments using Docker, Kubernetes, and CloudFormation, ensuring scalable and reliable deployment of data workflows. Adept at creating data visualizations and dashboards using Amazon QuickSight and .NET-based custom solutions for enhanced business intelligence.

Recognized for leading cross-functional initiatives, mentoring junior engineers, conducting technical reviews, and providing continuous support. Passionate about driving innovation through data engineering best practices, while fostering a collaborative and high-performing team environment.

TECHNICAL SKILLS:
- **Programming Languages:** SQL, Python, T-SQL, Java, Scala
- **Data Technologies:** Amazon Redshift, AWS Glue, Apache Kafka, Apache Airflow, Amazon EMR, Amazon Kinesis (for streaming), Apache Spark (PySpark), Hadoop (HDFS), Apache Hive, Apache HBase, AWS EMR, AWS Glue, DataLake, DataBrew, Azure Data Factory, Dataproc.
- **Cloud Platforms:** Amazon Web Services (AWS), Azure.
- **Containerization and Orchestration:** Docker, Kubernetes
- **Infrastructure as Code (IaC):** Terraform, AWS CloudFormation
- **Continuous Integration/Continuous Deployment (CI/CD):** Jenkins, AWS Code Pipeline
- **Version Control:** Git
- **ETL /WEB/BI Tools**: SSIS, SSRS, HTML, XML, IaaS.
- **Monitoring and Logging:** Amazon CloudWatch, Grafana
- **Database Systems:** Amazon RDS (for SQL Server, PostgreSQL)
- **Business Intelligence and Reporting:** Amazon QuickSight, Tableau
- **Machine Learning and Analytics:** Amazon SageMaker, AWS GPU Computing Technologies
- **Web Development Frameworks:** Play Framework, Akka
- **Messaging Systems:** Amazon MSK (Managed Apache Kafka)
- **Web Servers and Configuration:** Nginx
- **Data Governance and Catalog:** AWS Glue Data Catalog, AWS Lake Formation
- **Other Tools and Practices:** AWS CodeBuild, AWS CodeDeploy, AWS CodeCommit, AWS X-Ray (for A/B Testing), AWS Data Quality Assurance, Azure Storage Explorer.
- **Operating Systems:** Windows, Linux

PROFESSIONAL EXPERIENCE:
Lead Data Engineer | Anthem, Remote | September 2021- Present
Responsibilities:

- Developed ETL pipelines in and out of the data warehouse, creating major regulatory and financial reports using advanced SQL queries in Amazon Redshift.
- Implemented one-time data migration of multistate level data from SQL Server to Amazon Redshift using Python and T-SQL.
- Staged API or Kafka data (in JSON file format) into Amazon Redshift by flattening it for different functional services.
- Built Docker images to run Airflow on local environments for testing ingestion and ETL pipelines.
- Created and maintain Docker container clusters managed by Kubernetes for the CI/CD system to build, test, and deploy.
- Designed and managed scalable storage solutions using Azure Data Lake to support large-scale data processing and analytics.
- Working on Azure Synapse Analytics for implementing Pyspark Notebooks.
- Worked on Microsoft Azure Cloud to provide IaaS support to client.
- Developed Airflow DAGs for scheduling ingestions, ETL jobs, and various business reports.
- Supported the production environment and debug issues using Amazon CloudWatch logs.
- Loaded tables from Amazon S3 to Amazon Redshift for pushing them to Amazon Redshift.
- Written ETL jobs to read from web APIs using REST and HTTP calls, loading into Amazon S3 using .NET and AWS Glue.
- Created and maintained SSIS packages for ETL processes, including data extraction, transformation, and loading.
- Built scripts to load data from HDFS into Hive and contributed to ingesting data into the Data Warehouse using multiple data loading strategies.
- Worked closely with data engineers and architects to understand Hudi configurations and optimize data ingestion workflows for testing purposes.
- Employed Amazon Kinesis for real-time data streaming, enhancing the efficiency of data ingestion processes.
- Integrated Apache Airflow with Docker containers to orchestrate and schedule complex workflows efficiently.
- Utilized Git for version control and collaborated with a cross-functional team, ensuring smooth integration and code management.
- Performed ETL operations in Azure Databricks by connecting to different relational database source systems using JDBC connectors.
- Implemented AWS CloudFormation scripts for Infrastructure as Code (IaC) to automate and manage cloud infrastructure deployments.
- Build and Design ETL code to perform Data Integration and Data Transformation in BIDM- Business Intelligence Data mart using Informatica PowerCenter, Oracle and Unix.
- Leveraged AWS CodePipeline for continuous integration and continuous deployment (CI/CD) pipelines, ensuring rapid and reliable software delivery.
- Developed ETL workflows and data pipelines using Azure Databricks with PySpark to process high-volume data.
- Designed ETL workflows using Informatica to transfer data from flat files and Excel sheets into an Oracle Data Warehouse.
- Developed .NET-based data visualization dashboards for enhanced reporting.
- Integrated DynamoDB with other AWS services, such as Lambda, S3, and Redshift, to build comprehensive data pipelines and ETL processes.
- Built extract / load / transform (ETL) processes in the Snowflake Data Factory using dbt to manage and store data from internal and external sources.
- Developed scripts to transform and load structured and unstructured data into the data lake, improving accessibility and analytics capabilities.
- Applied best practices in data security by implementing encryption techniques for sensitive data at rest and in transit within the Amazon Redshift data platform.
- Implemented monitoring and alerting systems, utilizing tools such as Amazon CloudWatch and AWS CloudWatch to ensure the health and performance of deployed applications and infrastructure.
- Demonstrated proficiency in container orchestration and management using Amazon ECS, ensuring scalability and reliability of containerized applications.
- Implemented automated data quality checks to ensure accurate tracking and reporting for ad campaigns across multiple platforms.

Senior AWS Data Engineer | McKinsey, NY | March 2019- August 2021
Responsibilities:

- Orchestrated the seamless migration of an Oracle database to Amazon Redshift, significantly enhancing data accessibility and performance.
- Implemented Amazon QuickSight for robust reporting capabilities, contributing to improved data-driven decision-making.
- Led the data engineering efforts for media campaigns involving platforms like CM (Campaign Manager), SA (Search Ads), and DV360, ensuring efficient data integration and analysis.
- Conducted rigorous Unit and Integration testing for Informatica mappings to ensure data quality and compliance with business rules and objectives.
- Design and develop ETL process in AWS glue to migrate data from s3 into AWS redshift.
- Developed REST APIs in C# for seamless integration between Oracle HCM Cloud and custom applications deployed on Azure cloud.
- Implemented robust data governance, security protocols, and access control policies for data stored in Azure Data Lake.
- Created data pipelines for API and Redshift and developed python code for different tasks, dependencies for each job for workflow management and automation using Airflow tool.
- Collaborated with cross-functional teams to develop a framework for generating daily ad-hoc reports and extracts from enterprise data stored in Amazon Redshift.
- Led the design and deployment of Amazon EMR clusters and various Big Data analytic tools within the AWS platform.
- Configured and automated data integration workflows in SSIS, optimizing performance and ensuring data accuracy.
- Kept up-to-date with the latest trends and best practices in React.js and JavaScript development to continuously improve skills and apply new techniques.
- Conducted performance testing of Apache Hudi operations (ingestion, updates, and deletes) to ensure low-latency data processing in high-volume environments.
- Designed and implemented ETL processes in Informatica to load data from flat files and Excel sources into the Oracle Data Warehouse.
- Integrated data from various sources, including social media, search, and display ads, to create a unified view of customer interactions and media effectiveness.
- Collaborated on optimizing data transformations and building advanced analytics solutions in Databricks notebooks.
- Developed interactive and insightful dashboards in Amazon QuickSight, catering to specific business requirements and user needs for data analysis and decision-making.
- Migrated on-premises Hadoop systems to AWS, demonstrating proficiency in cloud platform transitions.
- Played a key role in designing and implementing advanced analytical models within Amazon EMR over extensive datasets, collaborating closely with the Data Science team.
- Developed scripts in Hive SQL for creating intricate tables with performance-centric features like partitioning, clustering, and skewing.
- Utilized AWS Glue and other AWS APIs for monitoring, querying, and billing analysis, contributing to enhanced Amazon Redshift usage efficiency.
- Implemented Amazon Redshift row-level security features and views to facilitate secure data sharing with other teams.
- Worked on tag management solutions to ensure consistent and efficient data collection across marketing platforms.
- Enhanced and debugged existing SSIS solutions to improve data processing efficiency and troubleshoot data quality issues.
- Provided Guidance to development team working on Pyspark as ETL platform.
- Implemented Java-based RESTful APIs using Spring Boot for data access and manipulation in AWS and Azure environments.
- Used SSIS to create ETL Packages to validate, extract, transform and load data to data warehouse databases and data marts.
- Developed automated test scripts to continuously validate Hudi functionalities, including incremental data processing and time travel features, enhancing testing efficiency.

- Leveraged GPU computing technologies on AWS for the automation of machine learning and analytics pipelines, contributing to enhanced scalability and performance.
- Implemented DataVault architecture in Azure Data Factory for handling complex data transformations and integrations in the Affordable Care Act-HCR data scrubs project.
- Collaborated with marketing analysts and data scientists to build predictive models that optimize campaign performance and ad spend.
- Conducted proof-of-concept (POC) initiatives for utilizing machine learning (ML) models and Amazon SageMaker for table quality analysis in batch processes.
- Implemented data processing solutions leveraging Azure Databricks' integration with Azure Data Lake and Data Factory.
- Collaborated with cross-functional teams to develop and deploy advanced analytical models within Amazon Redshift, showcasing expertise in big data analytics.

Big Data Engineer | Northern Trust- Financial, NY | May 2018- Feb 2019
Responsibilities:

- Designed and implemented a real-time data processing pipeline using Amazon Kinesis and Amazon EMR, enabling the ingestion, transformation, and analysis of streaming data for immediate insights and decision-making.
- Orchestrated the migration of on-premises big data infrastructure to AWS, leveraging services like Amazon EMR, Amazon S3, and Amazon Redshift for scalable and cost-effective data processing solutions.
- Developed an end-to-end data pipeline for a retail analytics platform, integrating data from various sources and generating visualizations using Amazon QuickSight for actionable business insights.
- Implemented an Amazon EMR cluster, optimizing resource allocation and fault tolerance for efficient processing of large volumes of data.
- Developed Rest API's using python with flask and done the integration of various data sources including java, spreadsheets and Text files.
- Collaborated with database administrators to implement physical data models, optimizing database performance and ensuring data security.
- Developed dashboards and reports for real-time media campaign tracking using tools like Tableau, Power BI, and Looker.
- Build data pipelines in airflow in GCP for ETL related jobs using different airflow operators.
- Involved in the development of an AWS Glue data catalog and metadata management solution for improved data transparency and accessibility.
- Developed python scripts and Grafana dashboards to automate the monitoring tasks and fire alerts if data anomalies are detected.
- Implemented a data lake architecture using Amazon S3 and Amazon Redshift, enabling the organization to store and analyze diverse data types from different sources in a centralized and scalable repository for advanced analytics and reporting.
- Designed and deployed a machine learning model training pipeline using Amazon SageMaker and Python for predictive analytics, enabling the organization to derive actionable insights and make data-driven decisions based on historical and real-time data.
- Collaborated with cross-functional teams to build a scalable and fault-tolerant data processing system using Amazon Kinesis Analytics for real-time analytics, enabling the organization to monitor and respond to critical business events in real time.
- Extensively leveraged ETL (Extract, Transform, Load) methodologies to orchestrate the efficient loading of data from Oracle databases and flat files into the Data Warehouse.
- Designed Kimball-based data marts in Azure SQL Data Warehouse, providing stakeholders with actionable insights and analytics capabilities.
- Optimized the performance and scalability of a data warehouse environment by fine-tuning complex SQL queries, indexing strategies, and partitioning techniques in Amazon Redshift, resulting in improved query execution times and enhanced data retrieval efficiency.
- Developed Databricks ETL pipelines using notebooks, Spark Data frames, SPARK SQL, and python scripting.
- Implemented a data governance framework to ensure data quality, security, and compliance within the big data ecosystem, establishing robust data lineage tracking, access controls, and data retention policies to uphold regulatory standards and privacy requirements.

Data Engineer | Centene - Healthcare, Saint Louis, MO | July 2016 – April 2018
Responsibilities:

- Developed and maintained ETL workflows within Azure Data Factory, ensuring efficient data movement and transformation across various sources and destinations.
- Created and managed data pipelines for integrating and synchronizing data using Apache Kafka, ensuring seamless data ingestion into HDFS and PostgreSQL.
- Implemented Change Data Capture (CDC) logic in Azure Synapse to perform incremental data loads, optimizing data processing efficiency.
- Designed and developed Python scripts for data validation in Databricks and automated processes using Azure Data Factory to ensure data accuracy and consistency.
- Worked with Spark to process data from SQL Server to Hadoop, leveraging PySpark for ETL operations and ensuring smooth data flow between systems.
- Utilized Kafka for real-time data ingestion and messaging, creating topics for system logs and ensuring high scalability and performance in data processing pipelines.

Data Analyst | Mary Kay, Addison TX | July 2014 - June 2016
Responsibilities:

- Assisted in data cleaning and validation processes to improve data accuracy by 25%, ensuring reliable and consistent data for analysis and reporting.
- Created interactive Tableau dashboards to visualize key performance indicators (KPIs) and deliver real-time insights to stakeholders for better decision-making.
- Performed exploratory data analysis (EDA) to identify trends, patterns, and anomalies, contributing to actionable insights for business strategies.
- Utilized data visualization tools like Tableau, Power BI, and Python libraries (Matplotlib, Seaborn) to create informative and visually appealing reports and charts.
- Supported the development and maintenance of automated ETL pipelines, ensuring clean and reliable data for ongoing analysis and reporting.